

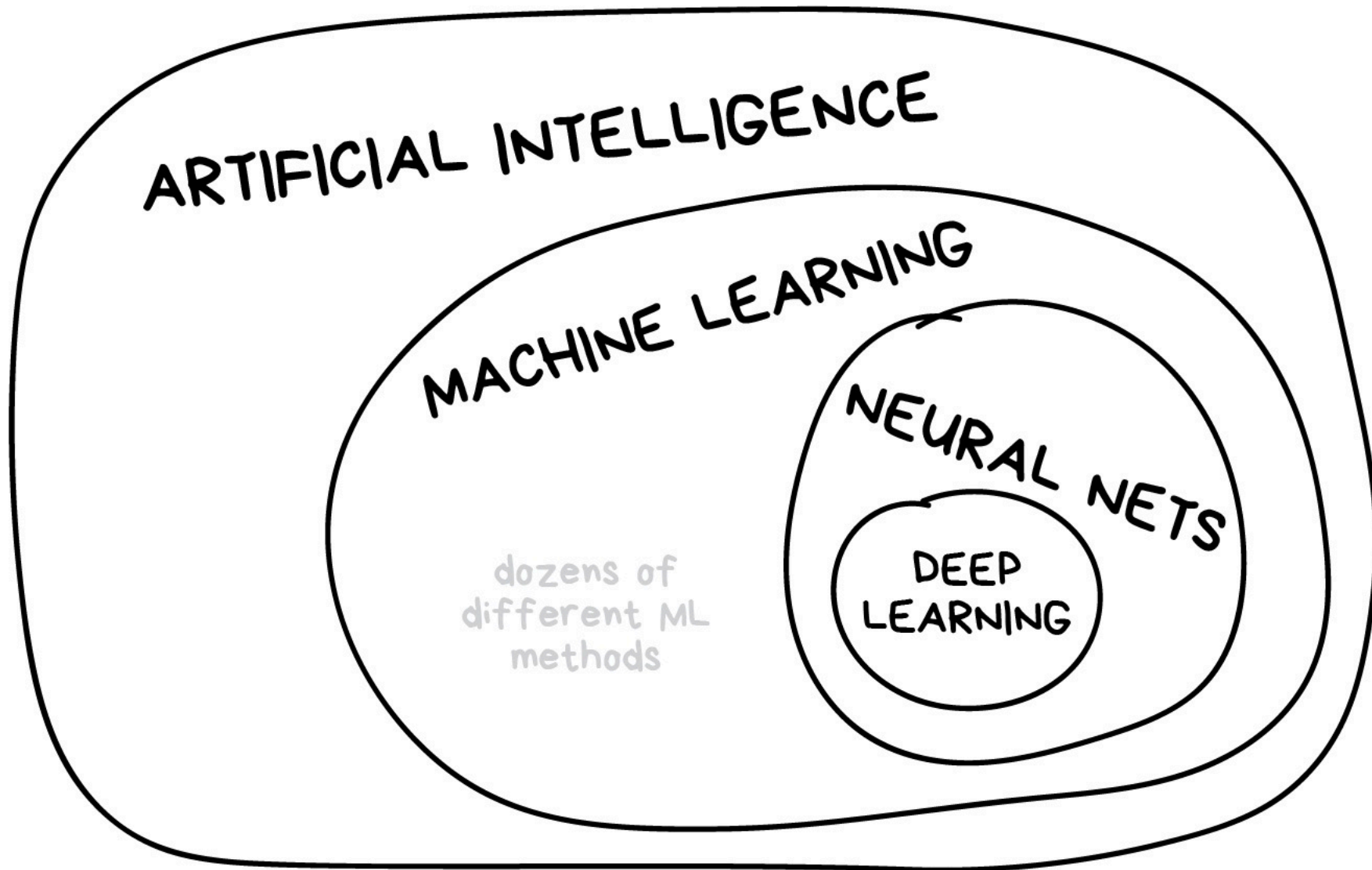
Intro To Modeling

Matthew McDonald

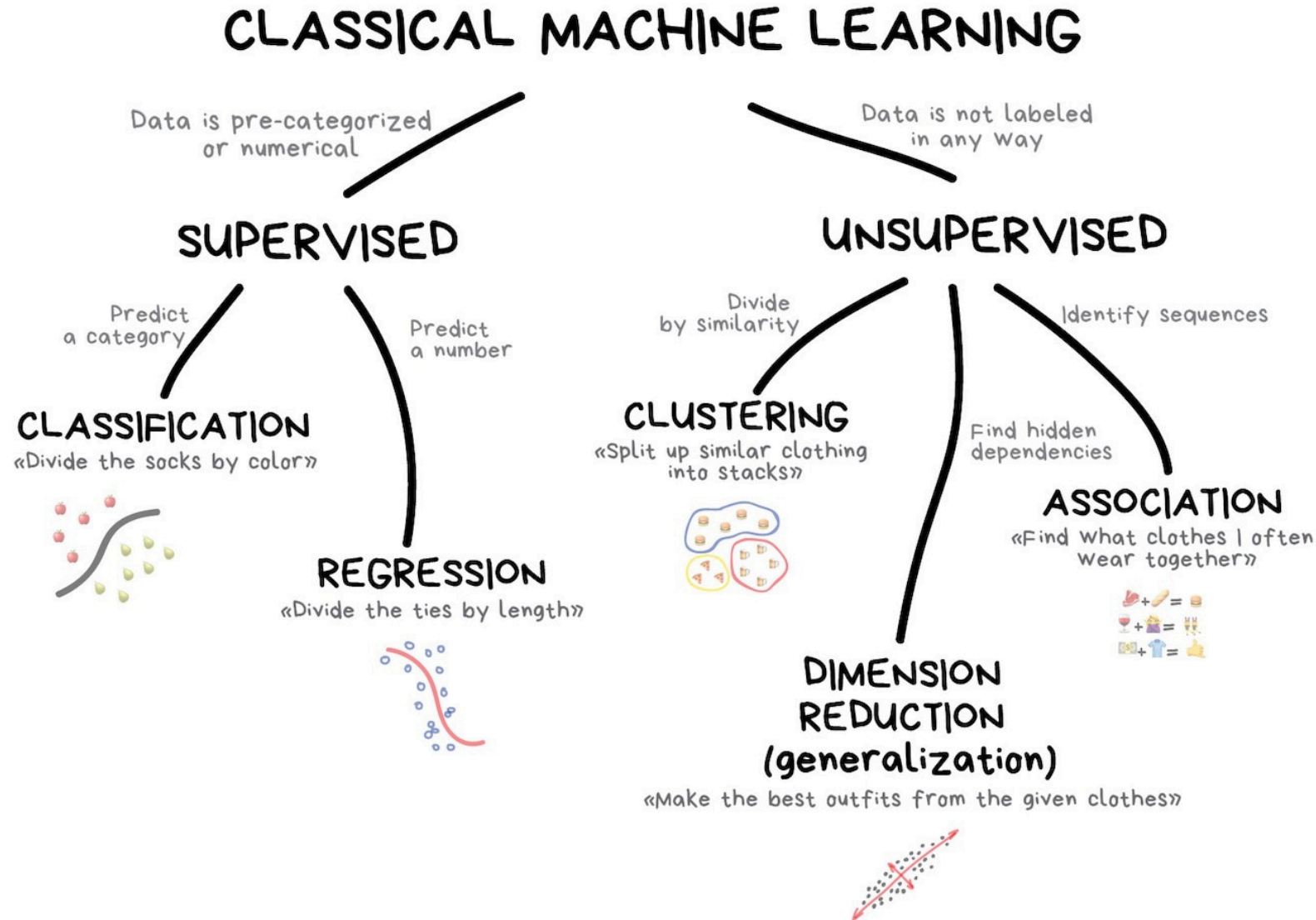
What is machine learning?



What is machine learning?



What is machine learning?



Why R For Modeling?

- R has cutting edge models.
 - Machine learning developers in some domains use R as their primary computing environment and their work often results in R packages.
- R and R packages are built by people who do data analysis.
- The S language is very mature.
- It is easy to port or link to other applications.
 - R doesn't try to be everything to everyone. If you prefer models implemented in C, C++, tensorflow, keras, python, stan, or Weka, you can access these applications without leaving R.
- The machine learning environment in R is extremely rich.

Downsides to Modeling in R

- R is a data analysis language and is not C or Java. If a high performance deployment is required, R can be treated like a prototyping language.
- R is mostly memory-bound. There are plenty of exceptions to this though.
- The main issue is one of consistency of interface. For example:
 - There are two methods for specifying what terms are in a model1. Not all models have both.
 - 99% of model functions automatically generate dummy variables.
 - Sparse matrices can be used (unless they can't).

The Modeling Process

Common steps during model building are:

- estimating model parameters (i.e. training models)
- determining the values of tuning parameters that cannot be directly calculated from the data
- model selection (within a model type) and model comparison (between types)
- calculating the performance of the final model that will generalize to new data

Many books and courses portray predictive modeling as a short sprint. A better analogy would be a marathon or campaign (depending on how hard the problem is).

What the Modeling Process Usually Looks Like

